



Cyber Security

Project 2

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM!

Step into the role of a Cybersecurity Analyst at DecodeLabs. Project 2 is your cryptographic phase: Basic Encryption & Decryption. This track isn't about "cracking codes"—it's about Data Confidentiality. Before you implement advanced security protocols, you must master the fundamental mechanics of obfuscating data and the logical reversible process of decryption. By completing this milestone, you are proving you can protect information in transit through pure mathematical and programmatic logic. Let's design a digital foundation that keeps data secure with absolute precision.



GUIDELINES:

Qualification Criteria

EARN YOUR BADGE 

- Requirement: Complete project 2 to unlock other projects for next week.
- Standard: All projects must be verified for quality.
- Mandatory: Project 2 is the vital bridge to professional development for every intern.



Project 2: Basic Encryption & Decryption

Goal:

Implement a simple encryption and decryption technique.

Key Requirements:

- Encrypt user text using a basic logic (Caesar cipher or similar)
 - Decrypt the encrypted text
 - Display both encrypted and decrypted output

Key Skills:

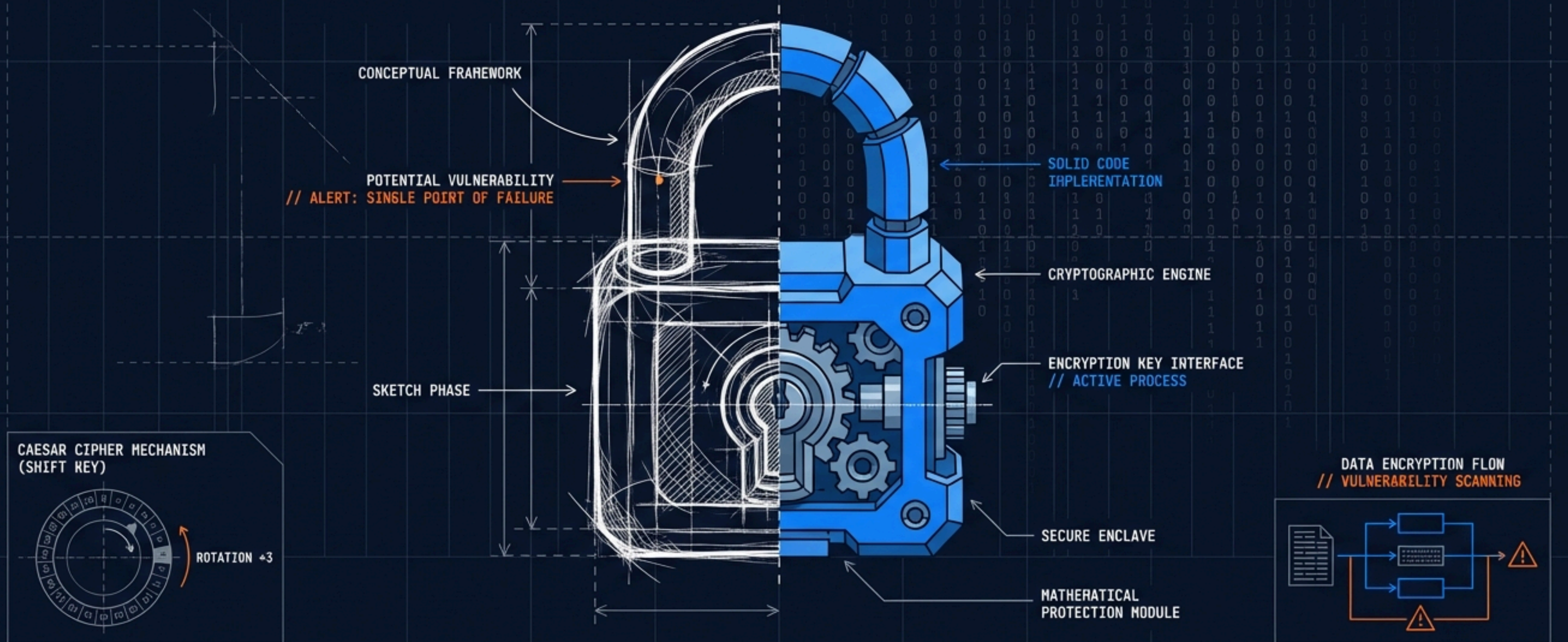
Encryption concepts, logic building, data protection basics



PROJECT 2: CONFIDENTIALITY LOGIC

The Architecture of Basic Encryption & The Caesar Cipher

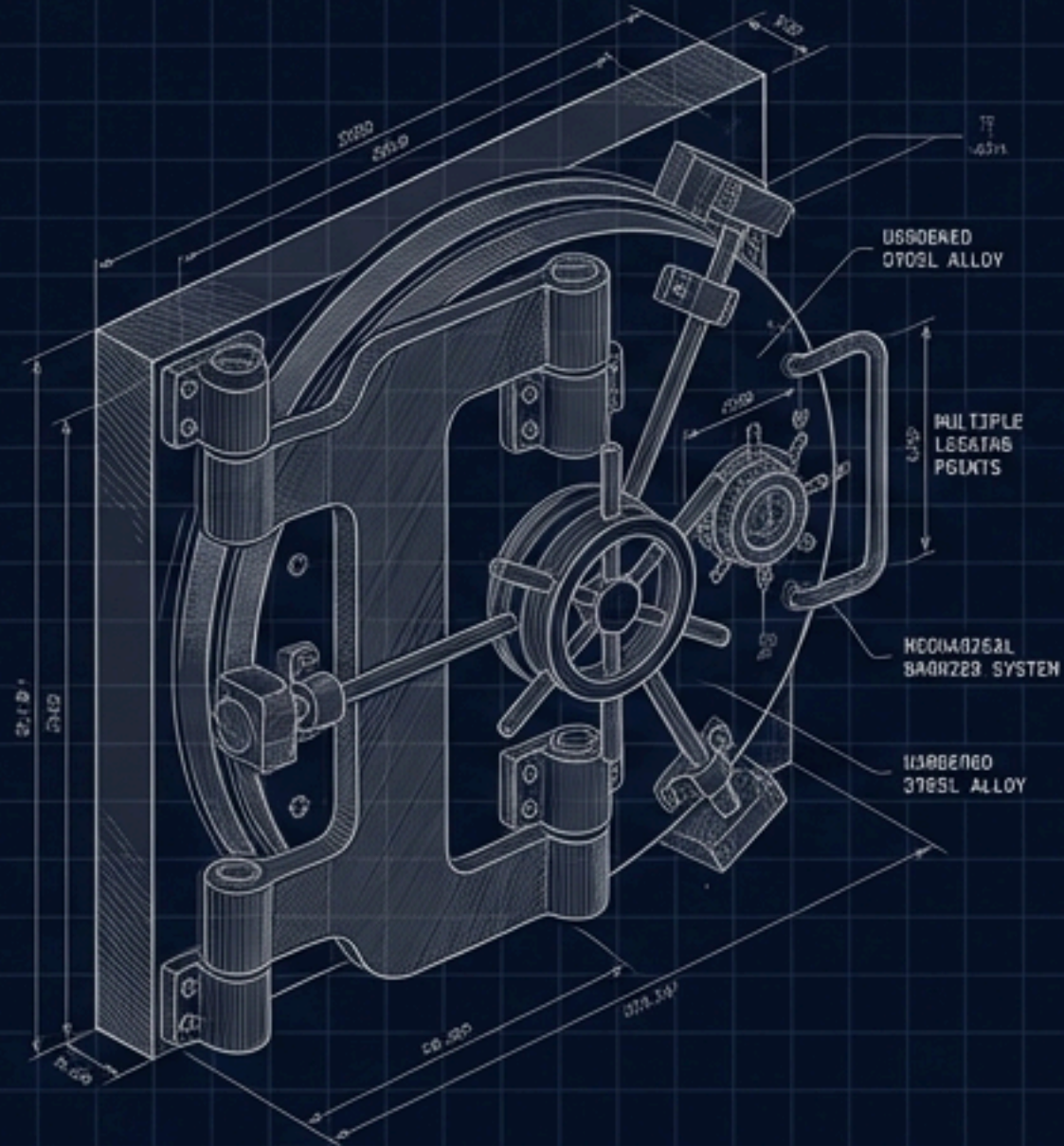
From Static Defense to Mathematical Protection



THE MISSION

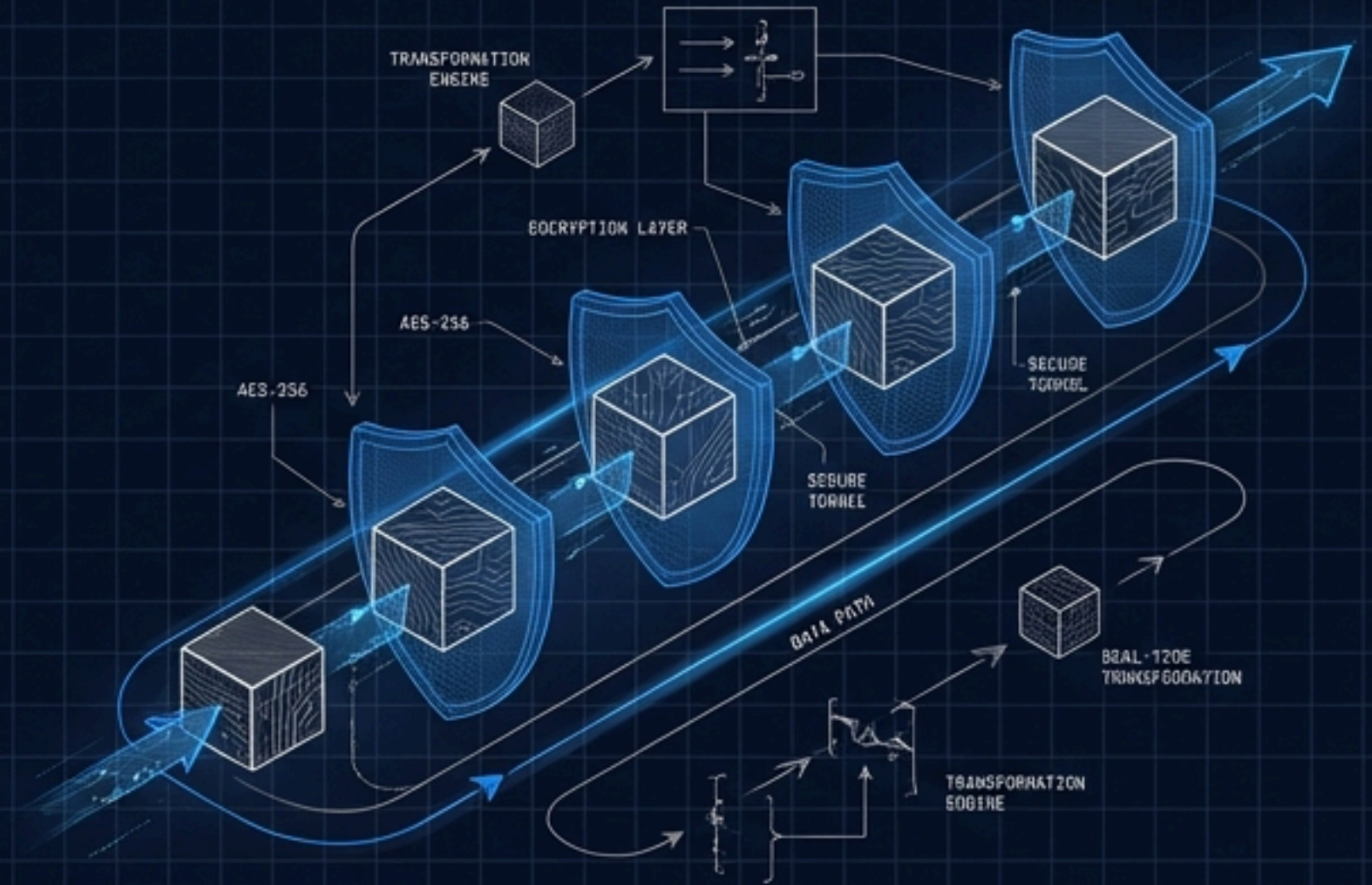
Protecting Data in Transit

STATIC DEFENSE



The Challenge: Data in motion cannot be protected by walls. It is exposed.

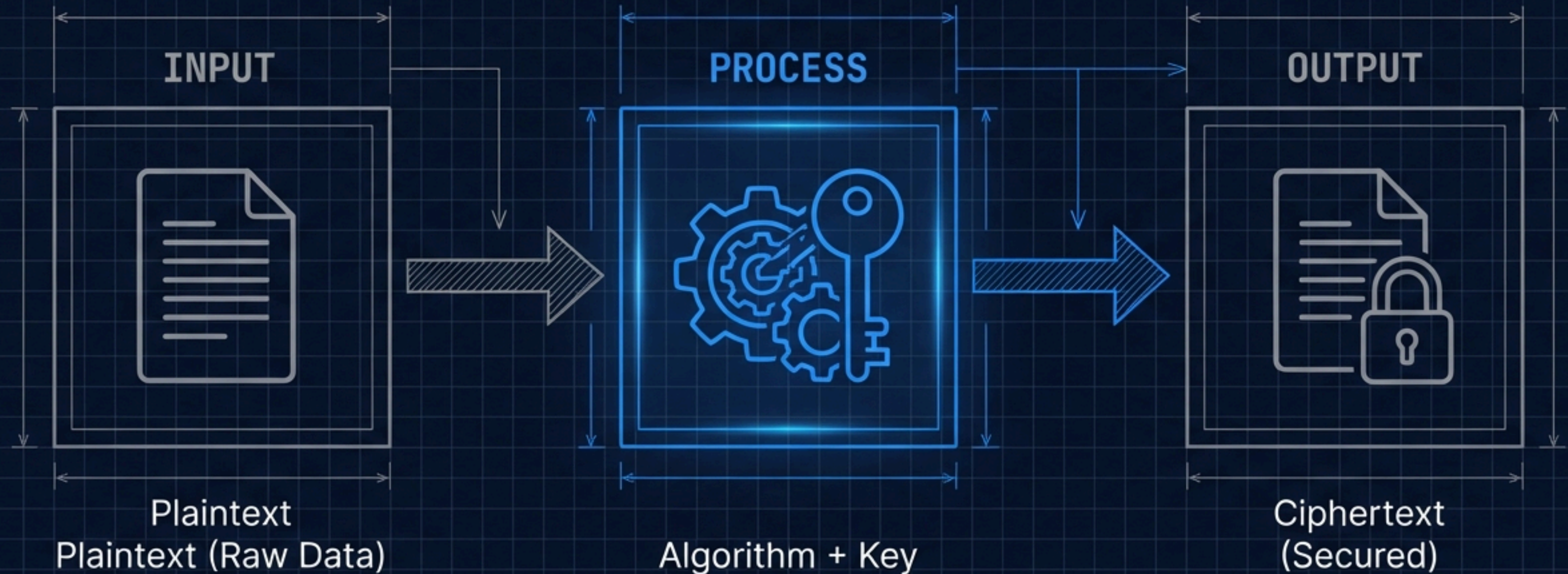
DYNAMIC PROTECTION



The Solution: Mathematical transformation.
Rendering data unreadable to unauthorized eyes.

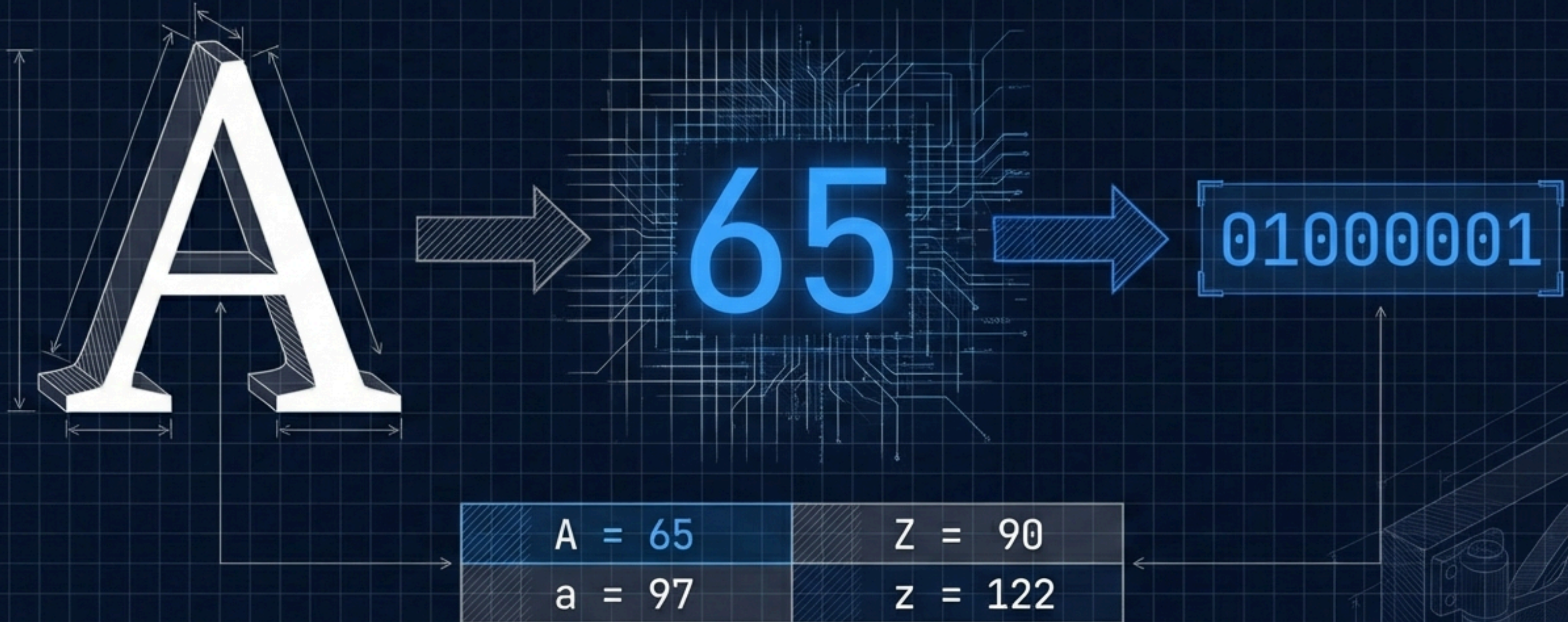
THE BLUEPRINT: IPO MODEL

The universal architecture of cryptographic systems.



INPUT LOGIC: ASCII

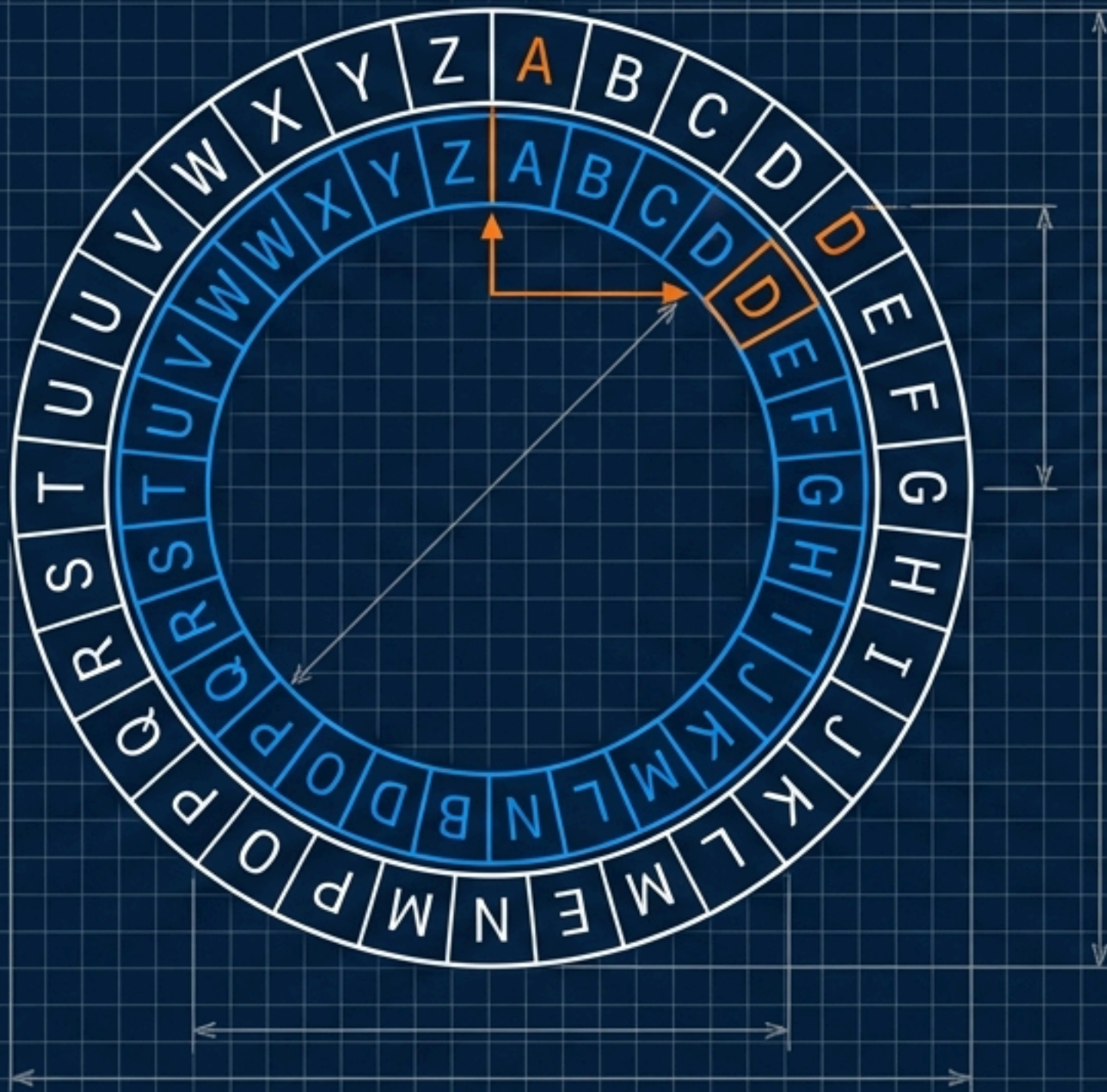
Text must become numbers before it can become math.



“We don’t shift letters; we shift integers.”

PROCESS LOGIC: THE SHIFT

Mono-alphabetic Substitution



$$E_n(x) = (x + n)$$

x = Character Position

n = Key (Shift)

THE MATH OF THE WRAP

Modular Arithmetic (%) handles the finite alphabet.

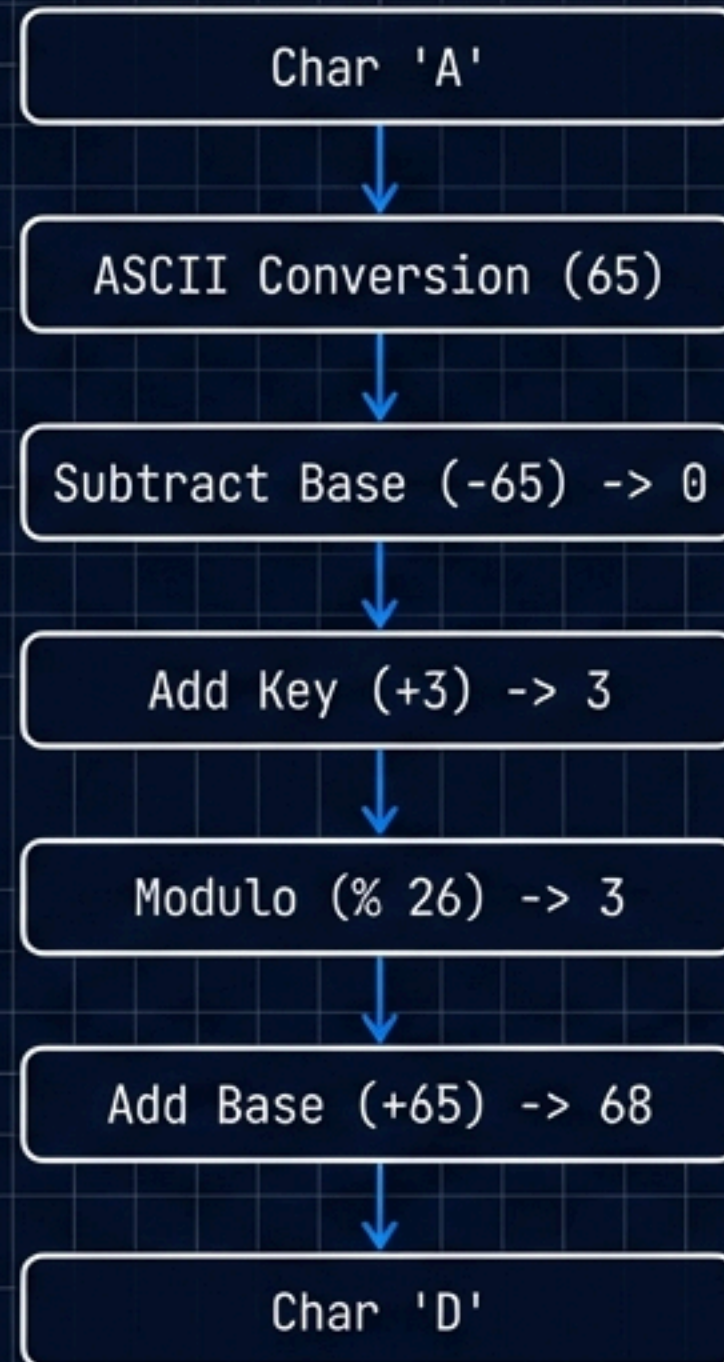


$$E_n(x) = (x + n) \% 26$$

Input **Y** (24) + Shift (3) = 27
27 % 26 = 1
1 = **B**

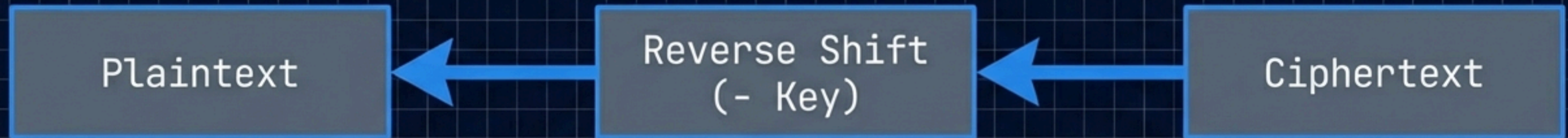
ALGORITHM VISUALIZATION

The step-by-step data flow for your code.



REVERSE ENGINEERING

Decryption Logic



$$D_n(x) = (x - n) \% 26$$

Symmetric Encryption: The same key locks and unlocks.

IMPLEMENTATION

Translating Math to Python

```
1  # Caesar Cipher Logic
2  shift = 3
3  char = 'A'
4
5  # The Critical Transformation
6  cipher_char = chr((ord(char) - 65 + shift) % 26 + 65)
```

Key Functions

ord() : Character to Integer

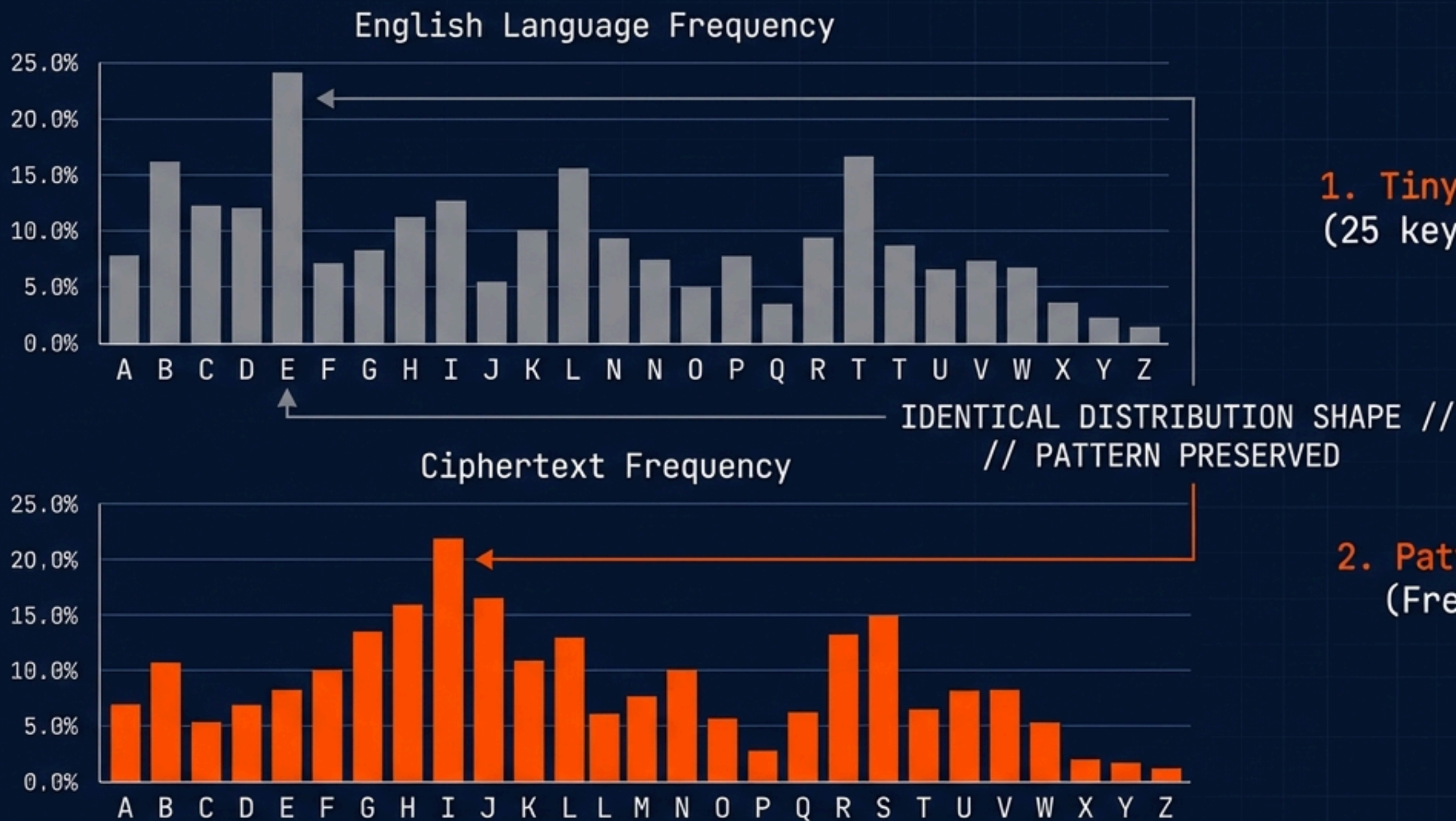
→

chr() : Integer to Character

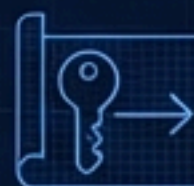
→

THE VULNERABILITY

Why the Caesar Cipher is a lockbox, not a vault.



1. Tiny Key Space
(25 keys = Instant Brute Force)

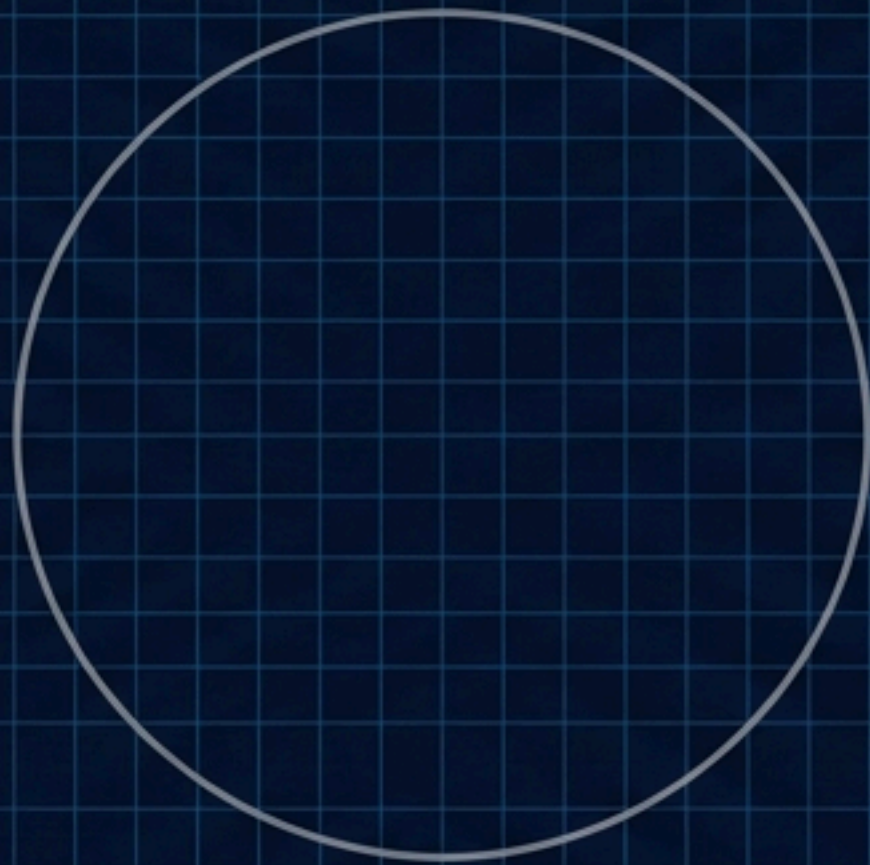


2. Pattern Preservation
(Frequency Analysis)

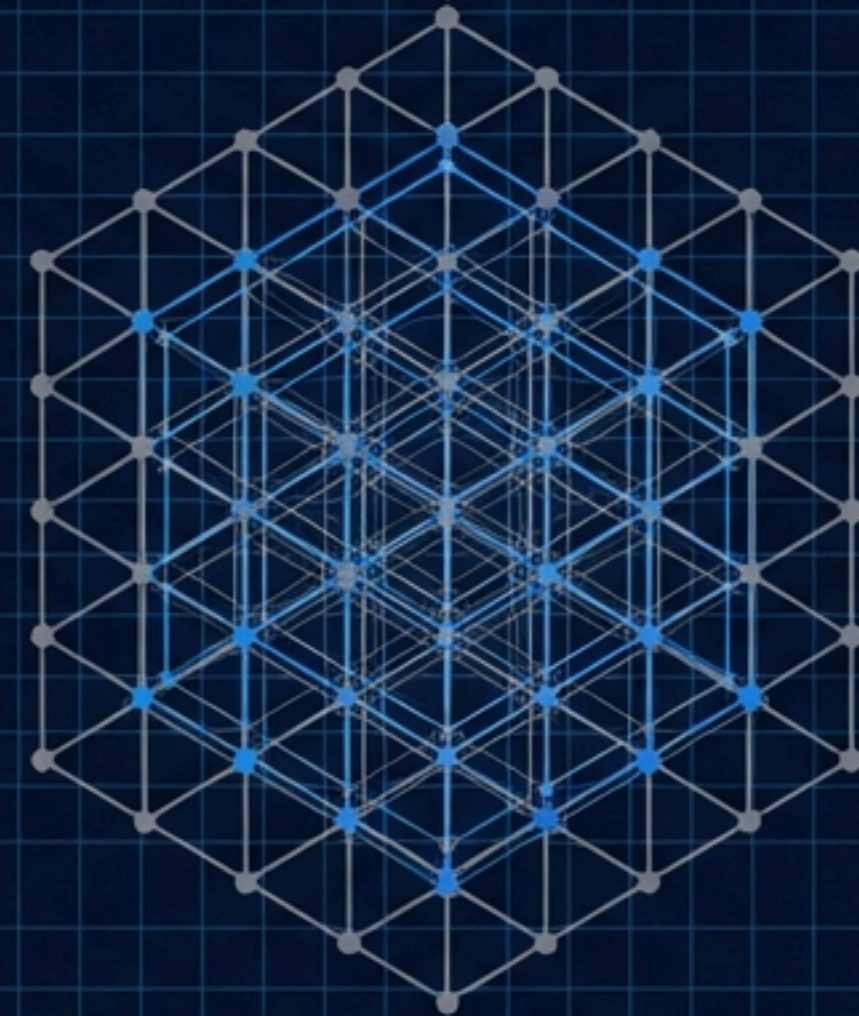


THE EVOLUTION

From Caesar to AES



Caesar Cipher



- Confusion + Diffusion
- 128-bit Keys
- XOR Operations

PROJECT DELIVERABLES

Build the Enigma

Inter

- ☐ Implement the IPO Cycle
- ☐ Apply the Math: `ord()`, `chr()`, `% 26`
- ☐ Handle Edge Cases (Spaces/Punctuation)
- ☐ Validate with Decryption Function

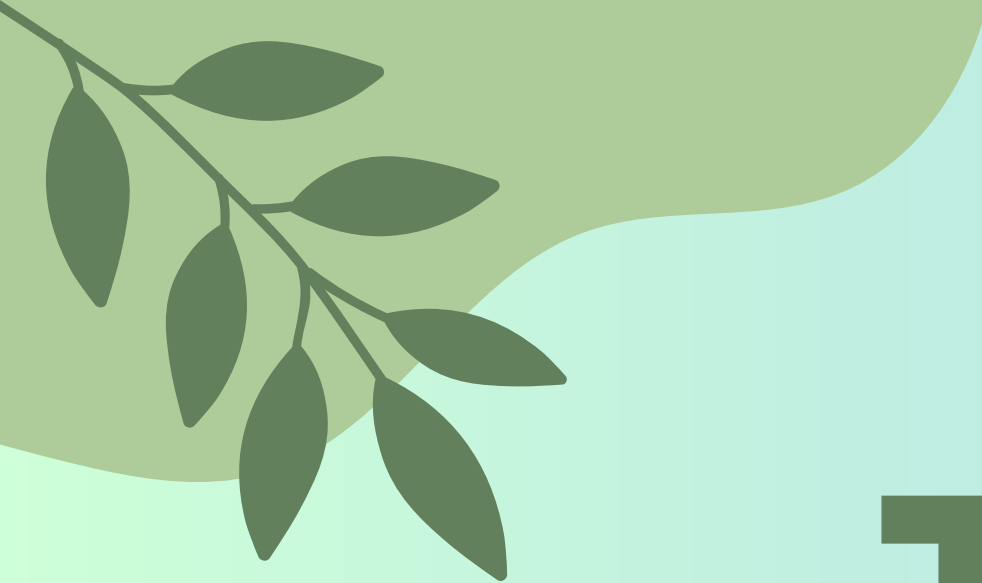
Caesar Cipher

- Confusion + Diffusion
- 128-bit Keys
- XOR Operations

Master the logic of the shift, and you master the foundation of the shield.

CONCLUSION

The absolute best way to master Cybersecurity is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions—like allowing the user to choose their own "shift key" or trying to implement a slightly more complex Vigenère cipher—and treat every logic error as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your technical depth to future employers. Your journey to becoming a professional security expert begins right here, right now, with the very first line of ciphertext you generate today.



THANK YOU

 +91 89330 06408

 decodelabs.tech@gmail.com

 www.decodelabs.tech

 GREATER LUCKNOW, INDIA

